

Optimization of Mechanical Thermal Energy Audit At Pelabuhan Ratu Unit 2 With Pi Vision

A. R. Yohanes*, Widayat¹, and Jaka Windarta¹

¹Magister Energy Departement, Diponegoro University, 50275 Semarang, Indonesia

Abstract. The percentage of coal PLTUs in Indonesia reached 45% of the capacity of 64,000 MW in 2018, and in the Java, Madura Bali (JAMALI) system it reached 74.84% of the total generating capacity of 46,958 MW in 2023. It is projected that coal PLTUs will continue to dominate the generation sector with a contribution of 50% of total generating capacity in 2050. The energy crisis is a strategic issue in current national development both in terms of price and availability. This is of course a result of the condition of world oil supplies which has an impact on domestic fuel conditions, considering that the majority of domestic fuel needs must be imported from other countries. This research aims to analyze the deviation of NPHR in the conditions of actual performance test results with commissioning, analyze the factors that cause an increase in NPHR, analyze the contributing contribution of each factor that causes an increase in NPHR by designing correction and reference factors related to the influence of changes in the main operating parameters of PLTU equipment on NPHR and look for energy saving opportunities and prepare recommendations for improving plant efficiency performance. This research was conducted at the coal-fired PLTU Pelabuhan Ratu Unit 2 which has an installed power capacity of 350 MW gross and is located in Sukabumi, West Java. In this research, an analysis of the boiler, turbine-generator, condensate, and feedwater systems was carried out. Actual data was obtained through a performance test procedure carried out under maximum load conditions of net capacity (DMN) for 2 hours in November 2022. Based on the research that has been carried out it can be concluded that there was an increase in NPHR during the actual performance test compared to the baseline of 72.38 kcal/ kwh, from 2,621.82 kcal/kwh to 2,694.20 kcal/kwh. The increase in NPHR has the potential to result in potential energy losses equivalent to IDR 15,365,850,000 per year. 3. Overall there are 7 goals for improving energy performance with improvements without investment, 4 improvements with low costs, and 4 improvements without costs. The total potential reduction in NPHR, if all recommendations are implemented, is 86.78 kcal/kWh or equivalent to a reduction in potential losses of IDR 44,792,896,565 per year with a total required repair cost of IDR 9,305,960,000. Keywords – PLTU, Energy Crisis, NPHR, Energy Efficiency.

1 Introduction

In Indonesia, electricity generation is still dominated by coal-fired steam power plants (PLTU). The percentage of coal PLTUs in Indonesia is 45% of the capacity of 64,000 MW in 2018, and in the Java, Madura Bali (JAMALI) system 74.84% of the total generating capacity of 46,958 MW in 2023. It is projected that coal PLTUs will continue to dominate the generation sector with a contribution of 50% of total generating capacity by 2050 [1]. Therefore, energy conservation at PLTUs must be improved as the dominant primary energy user until 2050. Systematic, planned, and integrated policies to conserve domestic energy resources and increase the efficiency of their use are known as energy conservation. To implement energy conservation is to use energy management, which is an integrated activity to control energy consumption so that it is used effectively and efficiently to achieve optimal output through structured and economical means [2].

The energy crisis is a strategic issue in current national development, in terms of price and availability. This is caused by the condition of oil supplies in the world, which has an impact on the conditions of domestic oil supplies because most of the domestic oil needs are imported from other countries. The country's oil reserves are running low because the heyday of Indonesian petroleum has passed. This refers to trends found in research conducted to predict Indonesia's petroleum production[3].

Previous research conducted studies on energy audits, including [4] conducting a detailed energy audit at a PLTU with a capacity of 210 MW, based on their research, the actual NPHR deviation value was obtained from the design at the same load. By comparing the actual operating parameter values with the design, we can determine the potential for improving plant efficiency performance. Equipment performance parameters, such as feedwater heaters and water preheaters, are not included in the deviations of the parameters studied. The parameter deviation values

* Corresponding author : A.r.yohanes@yahoo.com

between the actual values and the research design have not been calculated in NPHR units (kCal/kWh).

[5] conducted a study on the feasibility of establishing and calculating the costs of various energy inputs to and flows within a factory or organization in a certain period. Energy Audit of Coal Thermal Power Plants has been considered. The aim is to calculate all losses and provide steps to correct them and calculate the economic benefits after taking steps to correct losses. The study is based on the ASME Boiler Test Code method, Bureau of Energy Efficiency, and Guidelines for Thermal Power Plants (Indo German Energy Program). From the calculations, it can be easily concluded that the overall efficiency of the plant decreases with a decrease in the output load requirement. Load The thermal output of a power plant depends on the electricity demand. As the demand for electricity decreases, the output load of the thermal power plant decreases, and the overall efficiency of the plant is also lower because electricity cannot be stored so the plant runs at partial load, and the resulting output load overall efficiency of the plant is much higher.

[6] conducted an energy audit at PLTGU Block 1 Muara Karang, collected operating parameter data, and then calculated the performance of the main equipment. Calculations were compared with commissioning data in 1995. Based on the calculations, the energy profile was obtained, and the biggest gap from the comparison between actual performance and commissioning in 1995 analyzed the causes of equipment inefficiency carried out field studies and provided recommendations for Energy Saving Potential to increase equipment efficiency. These recommendations are analyzed with economic studies to provide alternative improvement solutions for management.

Several previous studies such as Improving Performance Through Energy Audits of 250 MW Coal Thermal Power Plants, Efficiency of Coal-Fired Boilers in Typical Thermal Power Plants[7], energy audits in power plants[8], Evaluation of Thermal Losses in 660 MW Coal Power Plants Using Efficiency Methods Indirect[9], Energy – Exergy Analysis of the Addition of SRO to the Dual Fuel Diesel Engine Exhaust System [10].

PLTU Pelabuhan Ratu is a generator that provides electrical power for the western part of Java. In carrying out its function as a provider of electric power, reliability, and efficiency are important both technically and economically. Therefore, to support the company's operational performance and sustainability, especially in terms of efficiency and operational activities, it is necessary to evaluate the energy use process in the form of an energy audit.

Based on the results of previous research studies, shows that there are still deficiencies in the implementation of energy audits that are not appropriate, so further research needs to be done on energy audits through various technologies such as PTMS, FEGT, DAT, and hybrid systems to obtain more valid and real-time data. Therefore, this research will focus on thermal and mechanical energy audits at PLTU Plabuhan Ratu Unit 2. To identify energy losses at coal PLTUs that have never been carried out, an energy audit

will be carried out by comparing the deviation of each operating parameter and main equipment performance between actual and baseline. Therefore, this research can be used as a guide for conducting energy audits at PLTUs, which can provide recommendations for improving energy performance based on the level of ease of implementation.

This research aims to analyze the deviation of NPHR in the conditions of actual performance test results with commissioning, analyze the factors that cause an increase in NPHR, analyze the contributing contribution of each factor that causes an increase in NPHR by designing correction and reference factors related to the influence of changes in the main operating parameters of PLTU equipment on NPHR and look for energy saving opportunities and prepare recommendations for improving plant efficiency performance.

2 Method

This research was conducted at the coal-fired PLTU Pelabuhan Ratu Unit 2 which has an installed power capacity of 350 MW gross and is located in Sukabumi, West Java. In this research, an analysis of the boiler, turbine-generator, condensate, and feedwater systems will be carried out. Actual data was obtained through a performance test procedure carried out under maximum load conditions of net capacity (DMN) for 2 hours in November 2022.

Data collection was carried out on the maximum load of DMN unit 2, 323 MW net. The load stabilization process and operating parameters when the net load of 323 MW is reached is 1 hour. After load and parameter stabilization are achieved, the duration of data collection for the performance test is 2 hours. Retrieval of data from the DCS data recorder at intervals of 15 minutes. The unit's load and operating parameters are kept stable during the performance test. Then, the value of each operating parameter during the performance test is set to match or approach the value during commissioning. Measurement of flue gas temperature and flue gas oxygen content at the inlet and outlet side of the Secondary Air Heater (SAH) and Primary Air Heater (PAH) using a portable flue gas analyzer. Using coal with the same High Heating Value (HHV) as that used during commissioning. Coal samples were taken from all coal feeders operating during the performance test 3 times, namely when the performance test had lasted for 30 minutes, 60 minutes, and 90 minutes. Then all coal samples obtained during the performance test are mixed evenly to obtain a new, representative sample for analysis in the laboratory. Coal is analyzed in the laboratory to obtain ultimate & proximate analysis data. Bottom ash samples were taken from the submerged chain conveyor in the amount of 1 kg 15 minutes before the performance test ended. The bottom ash sample will be analyzed in the laboratory to obtain unburnt carbon data on the bottom ash. Fly ash samples were taken from the electrostatic precipitator drain valve in the amount of 1 kg 15 minutes before the performance test ended. Fly ash samples will be analyzed in the laboratory to obtain data on unburnt carbon in fly ash. The location for measuring environmental temperature, humidity,

and atmospheric pressure data was taken near the boiler air fan using a portable temperature, humidity, and pressure meter. The blowdown valve and make-up water valve are closed during the performance test. Gross power is taken from the kWh meter output from the generator, while the net power and electricity itself are taken from the kWh meter output from the main transformer and auxiliary transformer in GIS.

The parameters observed in the boiler turbine generator system are Load gross, Main steam temperature & pressure, main steam flow, final feedwater temperature, feedwater flow, feedwater press, atemperator, condenser backpressure, Terminal Temperature Difference (TTD), auxiliary power, Final feedwater temperature, total coal flow, furnace inlet & outlet exit temp, O₂ content, coal parameters both proximate and ultimate. The maximum permitted load is 323 MW net. Data retrieval is only permitted under DMN maximum load conditions, because testing at other loads can cause disruption to network stability and reduce company revenue.

3 Result and Discussion

In this research, the analysis of energy performance calculations for generators and main equipment uses MS Excel software with the additional WinSteam add-in program. WinSteam is used to find water and steam property values such as enthalpy, entropy, density, specific volume, wetness level of saturated steam, etc. The equation for calculating the energy efficiency of plants and main equipment in MS Excel software refers to the ASME PTC standard. Equations should be centred and should be numbered with the number on the right-hand side.

3.1 Generetor Efficiency

Generator efficiency is the comparison between the generator output power and the generator input power. The generator input power is the same as the steam turbine output power. The turbine output power is obtained by multiplying the mass flow rate of steam by the difference in enthalpy of incoming and outgoing steam [11]. The steam turbine of PLTU Pelabuhan Ratu Unit 2 has 6 turbine extraction points. During the actual performance test, extractions 5 and 6 were not operated due to interference with the extraction supply valve, so the turbine power calculated was the power at each level of turbine extraction, namely extraction 1, 2, 3, 4, and turbine output. Temperature and steam pressure were obtained from measurement data, while extraction steam mass flow rate data was obtained from interpolation of the heat & mass balance diagram design at gross loads of 50%, 75%, and 100% capacity. The generator output power is taken from the kWh meter installed on the generator output side. Calculation data for turbine power and generator efficiency can be seen in Table 1.

Table 1. Calculation of Turbine Power and Generator Efficiency

Parameter	Unit	Aktual Value
Gross Generator Load	MW	297.12
Mass flow rate of incoming steam	ton/jam	844.51
Inlet steam pressure	MPa	14.50
Inlet steam temperature	°C	539.08
Inlet steam enthalpy	kJ/kg	3,424.59
Mass flow rate of extraction steam 1	ton/jam	47.51
Extraction steam pressure 1	MPa	5.49732705
Extraction steam temperature 1	°C	398.1624146
Extraction steam enthalpy 1	kJ/kg	3182.90
Mass flow rate of extraction steam 2	ton/jam	53.28
Extraction steam pressure 2	MPa	3.24249175
Extraction steam temperature 2	°C	330.5909297
Extraction steam enthalpy 2	kJ/kg	3063.77
Mass flow rate of extraction steam 3	ton/jam	35.39
Extraction steam pressure 3	MPa	1.543739582
Extraction steam temperature 3	°C	467.3297003
Extraction steam enthalpy 3	kJ/kg	3401.77
Extraction steam pressure 4	Mpa	0.81
Extraction steam temperature 4	°C	348.19
Extraction steam enthalpy 4	kJ/kg	3158.14
Mass flow rate of extraction steam 5	ton/jam	45.96
Extraction steam pressure 5	MPa	0.300
Extraction steam temperature 5	°C	266.14
Extraction steam enthalpy 5	kJ/kg	3000.726
Mass flow rate of exctraction steam 6	ton/jam	26.61
Mass flow rate of exctraction steam 4	ton/jam	51.82
Extraction steam pressure 6	MPa	0.05
Extraction steam temperature 6	°C	141.31
Extraction steam enthalpy 6	kJ/kg	375.21
Mass flow rate of exctraction steam 7	ton/jam	20.00
Extraction steam pressure 7	MPa	0.07
Extraction steam temperature 7	°C	96.39
Extraction steam enthalpy 7	kJ/kg	277.57
Mass flow rate of exctraction steam 8	ton/jam	1.61
Extraction steam pressure 8	MPa	0.01
Extraction steam temperature 8	°C	76.16
Extraction steam enthalpy 8	kJ/kg	2641.62
Power Turbine HP	MW	100.24
Power Turbine IP	MW	153.07
Power Turbine LP	MW	64.99
Total daya turbin	MW	317.07
Efficiency Generator	%	99.61

3.2 Data results for Own Electricity, GT Losses, and UAT Losses

Own electricity is the electrical power used by equipment in the generator. The loss in a transformer is the difference between the power entering the transformer and the power leaving the transformer. The kWh meter is only installed on the output side of the generator and GT so that the UAT input power and GT input power cannot be known, so a calculation is made of the total electricity usage itself, GT losses, and UAT losses by calculating the difference between the generator output power and the GT output power. Data and calculations of total electricity consumption, GT losses, and UAT losses can be seen in Table 2.

Table 2. Calculation of total electricity consumption, GT losses and UAT losses

Parameter	Unit	Actual value
Output generator power	MW	297.12
Output GT power	MW	286.53
Own electricity + GT Loss + UAT Loss	MW	14.18

Table 2 shows that the total electricity consumption, GT losses, and UAT losses were 14.18 MW, while the value at commissioning was 14.73 MW. There was a decrease in total self-consumption electricity, GT losses, and UAT losses of 0.55 MW.

3.3 Efficiency Boiler

Pelabuhan Ratu PLTU uses a subcritical, pulverized, and tangential firing type boiler. The design steam pressure entering the turbine at maximum load is 17.5 Mpa. A pulverizer or mill is used to grind coal into fine coal powder, making it easier for the coal to burn completely in the combustion chamber. The use of tangential firing technology at the Pelabuhan Ratu PLTU can also improve combustion quality because the combustion residence time increases and heat distribution in the combustion chamber is better when compared to front-rear firing technology. The efficiency calculation used in this research is the direct method and heat loss to determine the difference in the calculation results of the two methods. Based on the ASME PTC-46 standard, the NPHR value calculation uses the heat loss method as a basis for calculating boiler efficiency. The schematic diagram of the air and gas system at the Pelabuhan Ratu PLTU is shown in Figure 1.

Using the direct method shows the results of the boiler efficiency calculation during the actual performance test of 83.43%. The direct method cannot determine the source of the change in boiler efficiency that occurs because it does not calculate the magnitude of each type of heat loss in the boiler. Meanwhile, by using the heat loss method, the boiler efficiency is 82.24%. The results of the boiler efficiency calculation using the direct method are 1.19% higher than the heat loss method. The possible cause of the difference in calculation results is that the measurement of the coal feeder as a tool for measuring coal mass flow rate is

inaccurate. The accuracy of coal feeder measurements has a significant effect on the results of boiler efficiency calculations using the direct method.

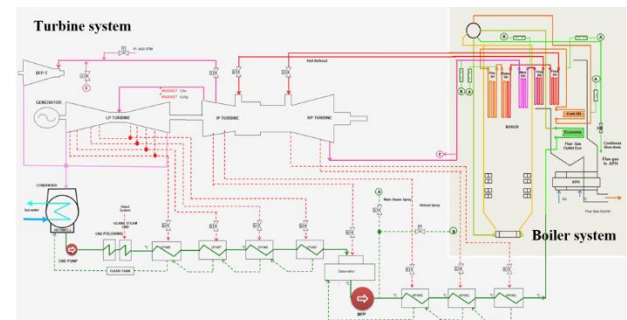


Fig. 1. Schematic diagram of the air and gas system in the PLTU Pelabuhan Ratu boiler

3.4 Isentropic Turbine Efficiency

Turbine efficiency calculations are carried out using the isentropic method where HP and IP turbine efficiency does not require steam flow rate variables, but only uses pressure and temperature measurements. While calculating LP turbine efficiency, exhaust loss must be considered because the amount is significant and the steam expansion process is in the saturation area

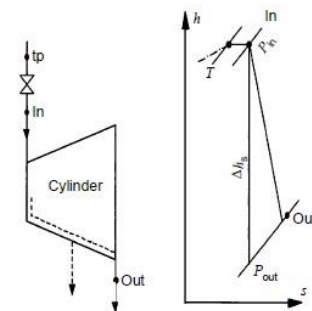


Fig. 2. Steam Expansion Process HP and IP Turbine steam expansion process

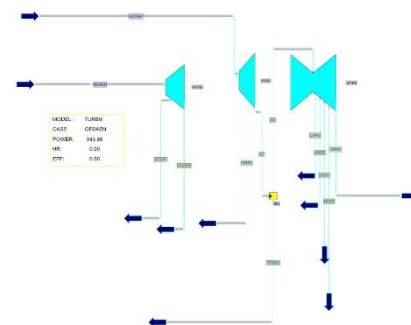


Fig. 3. Steam Power Plant turbine design

Based on calculations using gate cycle software, the isentropic efficiency of the HP Turbine is 88.3%, the IP Turbine is 91.3% and the LP Turbine is 67.5%.

Table 3. Relationship between equipment performance and losses heat rate

Parameter	Baseline	Actual	Faktor	Losses	kCal/kWh
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				De via si	Chang e	% HR	% HR	
1	Generator Gross Output (kW)	297,12	297,12	0,00				
2	NPHR Corrected (kCal/kWh)	2621,82	2694,20	72,38		2,76		72,38
	Subtotal (kCal/kWh)							24,99
	Other Factors (kCal/kWh)							47,39
A	Operator Controllable							
1	Flue Gas in AH, °C	351,97	372,25	20,28	5,50	0,35	1,29	33,84
2	Orsat O ₂ Gas in AH, %	4,44	4,35	-0,09	1,00	0,29	-0,02	-0,65
3	Main Steam Temperature, °C	540,00	539,08	-0,92	5,50	-0,15	0,03	0,66
4	Hot Reheat Steam Temperature, °C	540,00	536,08	-3,92	5,50	-0,14	0,10	2,62
5	Main Steam Pressure, kg/cm ²	16,45	14,50	-1,95	0,70	-0,04	0,11	2,92
6	Water spray to DeSH, %	5,94	12,91	6,97	1,00	0,03	0,17	1,20
7	Water spray to DeRH, %	3,01	0,85	-2,16	1,00	0,20	-0,43	-11,32
	Subtotal 1						2651,10	29,28
B	Unit Controllable							
1	Auxiliary Power, %	4,76	4,77	0,02	1,00	1,00	0,02	0,46
2	Final Temperature FW Heater	267,04	265,55	-1,49	2,50	-0,10	0,06	1,56
5	Condenser Vacuum, mmHgA	70,51	72,73	2,22	1,00	0,08	-0,17	-4,51
	Subtotal 2						2619,33	-2,49
C	Turbine Component							
1	HP Turbine Efficiency, %	85,93	85,18	-0,75	1,00	-0,18	0,13	3,52
2	IP Turbine Efficiency, %	91,54	94,66	3,12	1,00	-0,17	-0,53	-13,91
3	LP Turbine Efficiency, %	67,71	69,78	2,07	1,00	-0,45	-0,93	-24,46
4	BFPT-A Efficiency, %	60,00	54,76	-5,24	1,00	-0,02	0,10	2,75
5	BFPT-B Efficiency, %	60,00	57,71	-2,29	1,00	-0,02	0,05	1,20
	Subtotal 3						2590,92	-30,90
D	Cycle Component							
1	TTD HPH 1, °C	0,00	5,55	5,55	0,56	0,02	0,16	4,16
2	TTD HPH 2, °C	0,00	2,74	2,74	0,56	0,02	0,08	2,05
3	TTD HPH 3, °C	0,00	-2,18	-2,18	0,56	0,02	-0,06	-1,63
4	TTD LPH 5, °C	0,00	5,55	5,55	0,56	0,01	0,13	3,38
5	TTD LPH 6, °C	0,00	8,19	8,19	0,56	0,01	0,19	4,99
6	TTD LPH 7, °C	0,00	7,40	7,40	0,56	0,01	0,17	4,51
7	TTD LPH 8, °C	0,00	2	2,00	0,56	0,01	0,05	1,22
8	DCA HPH 1, °C	0,00	6,67	6,67	0,56	0,01	0,06	1,56
9	DCA HPH 2, °C	0,00	5,57	5,57	0,56	0,01	0,05	1,30
10	DCA HPH 3, °C	0,00	0,00	0,00	0,56	0,01	0,00	0,00
11	DCA LPH 5, °C	0,00	10,19	10,19	0,56	0,01	0,09	2,38
12	DCA LPH 6, °C	0,00	6,30	6,30	0,56	0,01	0,06	1,48
13	DCA LPH 7, °C	0,00	2,00	2,00	0,56	0,01	0,02	0,47
14	DCA LPH 8, °C	0,00	2,23	2,23	0,56	0,01	0,02	0,52
	Subtotal 4						2698,32	26,38
E	Boiler Component							

1	Moisture in Fuel (%)	29,55	38,51	8,96	1,00	0,10	0,90	23,50
2	H in Fuel (%)	3,00	3,02	0,02	1,00	2,00	0,04	1,05
3	Air Heater Leakage A, (%)	4,99	6,36	1,37	1,00	0,05	0,07	1,80
4	Air Heater Effectiveness A, (%)	67,22	68,97	1,75	1,00	-0,15	-0,26	-6,88
5	Air Heater Leakage B, (%)	4,99	14,31	9,33	1,00	0,05	0,47	12,23
6	Air Heater Effectiveness B, (%)	67,22	70,50	3,27	1,00	-0,15	-0,49	-12,88
7	FDF Inlet temperature, °C	29,00	31,44	2,44	5,50	-0,05	-0,02	-0,58
8	Air Temperature Outlet Mill, °C	65,00	62	-2,84	5,50	-0,04	0,02	0,54
9	Unburn Carbon, %	1,00	0,31	-0,69	1,00	1,00	-0,69	-17,98
F	Subtotal 5						2671,94	0,80
2	Make Up Water, (t/h)	6,00	6,82	0,82	1	0,09	0,07	1,92
	Subtotal 6						2671,14	1,92

Operator controllable are parameters that can be controlled by the operator while unit operation is in progress. This parameter is the easiest parameter to correct when there is a fairly high deviation so that the generating engineer can easily provide recommendations. The results of the NPHR deviation analysis due to heat losses to operators show that the total increase in NPHR caused by controllable operator factors is 29.28 kcal/kwh or the equivalent of an increase of 0.22% compared to commissioning results. This loss increased fuel costs of IDR 41,420,000 per day. The biggest loss for controllable operators is due to the decrease in flue gas temperature at the air heater inlet of 20.28 OC which increases NPHR of 33.84 kcal/kwh. The increase in temperature is caused by a decrease in heat transfer in the heat exchanger pipes in the boiler. The decrease in heat transfer in boiler pipes is generally caused by the presence of ash sticking to the surface of the boiler pipe. A decrease in steam temperature of 0.03 OC results in an increase in NPHR of 0.66 kcal/kwh. There was no significant change in steam temperature during the actual performance test.

The biggest loss in the controllable unit was due to an increase in condenser pressure of 9.89 mmHg. The increase in condenser pressure has an impact on increasing NPHR by 4.52 kCal/kWh or increasing fuel costs by IDR 6,390,00,000 per day. Based on technical report sources from EPRI (Palo Alto, 1998), there are several possible causes for increased condenser pressure, namely air leaks, increased condenser workload, condenser pipes covered with foreign objects, low cooling water flow rate, condenser pipes that are blocked due to leaks, the water box is not filled, the temperature of the cooling water entering the condenser increases, the performance of the steam ejector or vacuum pump decreases, and the performance of the cooling tower decreases.

Based on the results of NPHR deviation analysis between actual performance test values and commissioning, 19 factors were identified that caused the increase in NPHR at the Pelabuhan Ratu PLTU. The factors causing the increase in NPHR are opportunities to increase energy performance at the Pelabuhan Ratu PLTU. Priority for improving energy performance is

based on Pareto heat rate loss analysis with the principle of the 80/20 rule. This method shows that 80% of the increase in NPHR is caused by 20% of the causal factors that have been identified. The heat loss waterfall of the Pelabuhan Ratu PLTU can be seen in Figure 4.

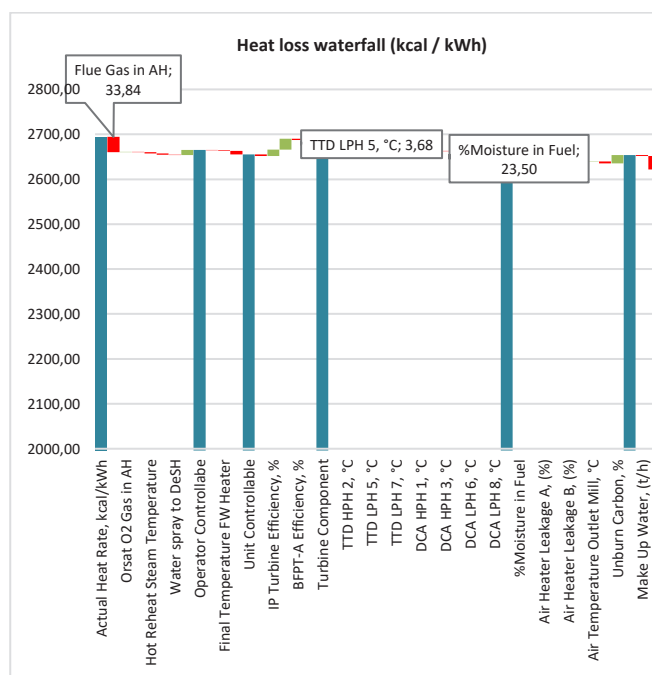


Fig. 4. Heat Loss Waterfall

Based on Figure 4, the increase in NPHR of 72.38 Kcal/Kwh was caused by an increase in flue gas temperature in the air heater, an increase in moisture content in fuel, an increase in air heater leakage B, an increase in TTD of HPH 1, an increase in TTD of LPH 6 and 7, and a decrease in outlet temperature. mill. Recommendations for improving energy performance were obtained from an analysis of the possible causes of each factor that resulted in an increase in NPHR based on references to the EPRI technical report (Palo Alto, 1998) and interviews from the PLTU XYZ engineering team. The average increase in PLTU NPHR if the plant is operated and maintained properly according to the manual is 2% every 10 years.

4 Conclusion

Based on the research that has been carried out, it can be concluded that there was an increase in NPHR during the actual performance test compared to the baseline of 72.38 kcal/kwh, from 2,621.82 kcal/kwh to 2,694.20 kcal/kwh. The increase in NPHR has the potential to result in potential energy losses equivalent to IDR 15,365,850,000 per year. 3. Overall there are 7 goals for improving energy performance with improvements without investment, 4 improvements with low costs, and 4 improvements without costs. The total potential reduction in NPHR, if all recommendations are implemented, is 86.78 kcal/kWh or equivalent to a reduction in potential losses of IDR 44,792,896,565 per year with a total required repair cost of IDR 9,305,960,000.

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